

Multiple Choice (10 marks)

1. Calculate the Larmor frequency for a proton in a magnetic field of 1 T.
 - (a) 23.56 GHz
 - (b) 50 MHz
 - (c) 100 MHz
 - (d) 42.58 MHz
 - (e) 1 GHz
2. Calculate the normality of a solution 40 ml of which is required to neutralize 0.56 g of KOH.
 - (a) 0.5 N
 - (b) 0.2 N
 - (c) 0.75 N
 - (d) 0.1 N
 - (e) 1 N
3. Each resonance form of the nitrate ion, NO_3^- , has how many sigma and how many pi bonds?
 - (a) 4 sigma and 1 pi
 - (b) 3 sigma and 1 pi
 - (c) 0 sigma and 4 pi
 - (d) 3 sigma and 2 pi
 - (e) 2 sigma and 1 pi
4. In a chemical reaction requiring two atoms of phosphorus for five atoms of oxygen, how many grams of oxygen are required by 3.10 g of phosphorus?
 - (a) 9.30 g O
 - (b) 2.50 g O
 - (c) 4.00 g O
 - (d) 5.00 g O
 - (e) 3.10 g O
5. A sample of 255mg of neon occupies 3.00dm^3 at 122K. Use the perfect gas law to calculate the pressure of the gas.

- (a) 4.20×10^{-2} atm
- (b) 5.55×10^{-2} atm
- (c) 6.33×10^{-2} atm
- (d) 3.15×10^{-2} atm
- (e) 8.21×10^{-2} atm

True / False (10 marks)

Answer True or False

6. True or False: The magnetogyric ratio of ^{23}Na is $9.45 \times 10^7 \text{ T}^{-1} \text{ s}^{-1}$, which is commonly used for high-resolution NMR studies.
7. True or False: When a solid melts, there is a positive enthalpy change and a positive entropy change due to increased disorder.
8. True or False: The molar weight of the non-electrolyte solute can be determined as 130 g/mol by using the boiling point elevation method.
9. True or False: Induced dipole interactions are generally stronger than dipole-dipole interactions in most molecular scenarios.
10. True or False: The energy of one mole of UV photons with a wavelength of 300 nm is approximately 450 kJ/mol.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Analyze the deposition of zinc in an electrolytic cell when a current of 5.36 A is applied for 30 minutes, resulting in 3.27 g of zinc. Calculate the equivalent weight of zinc and discuss its significance in electrochemistry.
12. Calculate the voltage of the cell $\text{Pt}/\text{H}_2/\text{HCl} (0.2 \text{ M})/\text{HCl} (0.8 \text{ M})/\text{H}_2/\text{Pt}$ at 25°C. *given that $t_{\text{H}^+} = 0.83$. Discuss the underlying principles and calculations involved in your answer.*

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